$E(n)$-POWER OPERATIONS AND ISOGENIES

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Abstract. These notes are for the talks at AMSS during 2026 spring. We discuss topics related to $E(n)$-power operations, their algebraic approximations, congruence criteria, the associated isogenies and more.

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Through these talks, we fix a characteristic $p > 0$ and a height $n > 0$. We set $E = E(n)$ to be the Morava E-theory associated with a height n formal group $\mathbb{G}$, $K = K(n)$ to be the Morava K-theory and use $\mathfrak{m}$ to denote its maximal ideal.

Suppose we have a bundle $W \to X$, we will use $S(W)$ to denote its sphere bundle and X^W to denote the Thom space of this bundle.

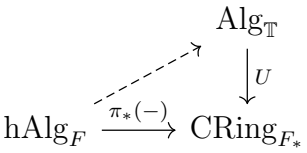
1. Algebraic approximation

Let F be a commutative ring spectrum, by commutative, we mean $\mathbb{E}_\infty$ or $\mathbb{H}_\infty$. The power operations are natural transformations $\pi_i(-) \to \pi_j(-)$ of functors $\mathrm{hAlg}_F \to \mathrm{Sets}$. In hAlg_F, π_i is corepresented by $F\{t\} = \mathbb{P}(\Sigma^i F) \in \mathrm{hAlg}_F$, where $\mathbb{P}(-) = \bigoplus_{m\geqslant 0} \mathbb{P}_m(-)$ is the free commutative F-Alg monad on hMod_F. By Yoneda, one has the following bijection between sets.

$$\{\text{Power operations } \theta : \pi_i(A) \to \pi_j(A)\} \longleftrightarrow \pi_j(\mathbb{P}(\Sigma^i E))$$

This implies that for $R \in \mathrm{hAlg}_F$, $\pi_*(R)$ is not only an F_*-algebra. It has other structures given by the power operations.

Roughly speaking, by an *algebraic approximation of power operations* we mean a "package" which contains these additional structures of $\pi_*(R)$ in a purely algebraic sense. More precisely, it is a monad $\mathbb{T}$ that acts on Mod_{F_*}, whose algebra contains almost all the data of power operations. It should fit into the following picture.



For $F = E(n)$ the Morava E-theory, this is the monad $\mathbb{T}$ defined by [Rez09], which is the main object we are going to discuss.

1.1. **Construction of $\mathbb{T}$.** We use $\mathrm{Mod}_E^{\mathrm{ff}}$ to denote the full subcategory of Mod_E generated by finite free objects. The objects of $\mathrm{Mod}_E^{\mathrm{ff}}$ are of the form $\bigvee_{i=1}^m \Sigma^{d_i} E$. It is easy to see that $\mathrm{Mod}_E^{\mathrm{ff}}$ is stable under retract and finite sum. The functor $\pi_*(-)$ induces an equivalence $\mathrm{hMod}_E^{\mathrm{ff}} \simeq \mathrm{Mod}_{E_*}^{\mathrm{ff}}$.

The construction idea of $\mathbb{T}$ is to Kan-extend some suitable functor $F\colon \mathrm{Mod}_{E_*}^{\mathrm{ff}} \simeq \mathrm{hMod}_E^{\mathrm{ff}} \to \mathrm{Mod}_E$ along the inclusion $\mathrm{Mod}_{E_*}^{\mathrm{ff}} \to \mathrm{Mod}_{E_*}$. The first candidate for F is $\pi_*(\mathbb{P}_m(-))$, which is intuitively an algebraic counterpart of the free commutative algebra monad. However, this is not the correct choice since this does not give the correct dual. The correct choice is $\pi_*(L\mathbb{P}_m(-))$ as shown in [HS99], where L is the $K(n)$-localization functor. It is equivalent to $T(n)$-localization or $\mathfrak{m}$-completion when restricting to E-modules as discussed in [HS98].

The following proposition points out the advantage of this choice.

Proposition 1.1. *Let L denote the $K(n)$-localization functor. Then $L\mathbb{P}_m(-)\colon \mathrm{Mod}_E \to \mathrm{Mod}_E$ will send finite free E-modules to finite free E-modules.*

The proof of Proposition 1.1 is given by combining two lemmas.

Lemma 1.2. *Let ρ_p denote the regular representation of C_p and let $c \in \mathbb{Z}$. Then $E_*^\wedge(BC_p^{c\rho_p})$ is finitely generated and free.*

Proof. We begin with the case where $c = 0$. Let λ denote the universal line bundle over BS^1. Then we have the following cofiber sequence:

$$S(\lambda^p) \to BS^1 \to (BS^1)^{\lambda^p}$$

Applying E-cohomology, together with Thom isomorphism, we have the Gysin sequence:

$$0 \to E_*[\![t]\!] \xrightarrow{z} E_*[\![t]\!] \to E^*(S(\lambda^p)) \to 0$$

The map z sends the Thom class (or complex orientation) to the Euler class (or the top Chern class). Therefore, t is sent to $[p]t$, which equals to t^{p^n} after mod $\mathfrak{m}$. This implies that $E^*(S(\lambda^p)) \cong E^*(BC_p)$ is finitely generated free with generators $1, t, \cdots t^{p^n-1}$. The the dual $E_*^\wedge(BC_p)$ is finitely generated free.

In general, suppose that c is even, then $c\rho_p = \frac{c}{2}\rho_p \otimes \mathbb{C}$ admits a complex structure. Using Thom isomorphism, we have $E_*^\wedge(BC_p^{c\rho_p}) \cong E_*^\wedge(BC_p)$ to be finitely generated free.

Finally, suppose that c is odd. By Thom isomorphism again, we reduce to the case where $c = 1$. Then we have a split $\rho_p \cong \mathbb{R} \oplus \overline{\rho}_p$. If p is odd, then $\overline{\rho}_p$ has a complex structure, then we conclude from Thom isomorphism. If $p = 2$, then $\overline{\rho}$ is simply the sign representation. Then in this case, $E_*^\wedge(BC_p^{\rho_p}) \cong E_*^\wedge(\Sigma BC_p)$ is finitely generated free. □

Once we have this first lemma, we can deduce immediately that $L\mathbb{P}_{C_p}$ sends finite free rank one E-modules to finite free E-modules. In order to generalize to arbitrary Σ_m and M, we need the second lemma.

Lemma 1.3. *Let $G \subset \Sigma_m$ contains a p-Sylow subgroup. Then $\mathbb{P}_G(M) \to \mathbb{P}_m(M)$ admits a section.*

Proof. For every $X \in \mathrm{Sp}$, we have the following diagram.

$$\begin{array}{ccccc}
\mathrm{Map}_{\mathrm{Sp}}(X, \mathbb{P}_G(M)) & \xrightarrow{\simeq} & \mathrm{Map}_{\mathrm{Sp}^G}(X, M^{\otimes m}) & \xrightarrow{\simeq} & \mathrm{Map}_{\mathrm{Sp}^{\Sigma_m}}(\Sigma^\infty_+ \Sigma_m/G \wedge X, M^{\otimes m}) \\
\downarrow & & & & \simeq \downarrow \\
 & & & & \mathrm{Map}_{\mathrm{Sp}^{\Sigma_m}}((\Sigma^\infty_+ \Sigma_m/G)_{(p)} \wedge X_{(p)}, M^{\otimes m}) \\
 & & & & \downarrow \\
\mathrm{Map}_{\mathrm{Sp}}(X, \mathbb{P}_m(M)) & \xrightarrow{\simeq} & \mathrm{Map}_{\mathrm{Sp}^{\Sigma_m}}(X, M^{\otimes m}) & \xrightarrow{\simeq} & \mathrm{Map}_{\mathrm{Sp}^{\Sigma_m}}((\Sigma^\infty_+ \Sigma_m/\Sigma_m)_{(p)} \wedge X_{(p)}, M^{\otimes m})
\end{array}$$

The first top and bottom horizontal equivalences are Adams isomorphisms. The right vertical map is induced by a retraction for $\Sigma^\infty_+ \Sigma_m/G \to \Sigma^\infty_+ \Sigma_m/\Sigma_m$ by hypothesis. □

Proof of Proposition 1.1. We have an equivalence $(S^c)^{\wedge p}_{hC_p} \simeq BC_p^{c\rho_p}$ (when $c = 0$, both sides are BC_p, when $c \neq 0$, they have the same cell structure). Then Lemma 1.1 implies immediately that $\pi_*(L\mathbb{P}_{C_p}(\Sigma^c E))$ is finitely generated free. By Lemma 1.3, this implies that $\pi_*(L\mathbb{P}_p(\Sigma^c E))$, which is a retract of $\pi_*(L\mathbb{P}_{C_p}(\Sigma^c E))$, is finitely generated free.

For general M, N, we have the following binomial formula:

$$\mathbb{P}_p(M \vee N) = \bigvee_{i+j=p} \mathbb{P}_i(M) \wedge \mathbb{P}_j(N)$$

Each i (or j) is smaller than p, then by Lemma 1.3 again, we see that $\pi_*(L\mathbb{P}_{C_p}(\Sigma^c M))$, is finitely generated free for every finite free M.

Finally, by [Bru+86, Proposition I.2.5], L preserves any cohomology equivalence. Therefore, we have $L\mathbb{P}_{\Sigma_{p^{lr}}} = L\mathbb{P}_p \cdots \mathbb{P}_p = L\mathbb{P}_p L\mathbb{P}_p \cdots L\mathbb{P}_p$, which takes finite free modules to finite free modules. Again using Lemma 1.3, we see that $\mathbb{P}_m$ takes finite free modules to finite free modules. □

The above proof in fact shows more. For example, we have:

Corollary 1.4. *Suppose that c is odd, then $\pi_*(L\mathbb{P}_p(\Sigma^c E))$ is odd.*

Corollary 1.5. *For $A \in \mathrm{CAlg}^\wedge_E$ and any odd element $x \in \pi_q(A)$, we have $x^2 = 0$.*

Proof. The element x is represented by $f\colon \Sigma^q E \to A$. Then x^2 is the image of some element in $\pi_{2q}(L\mathbb{P}_2 E)$ var $L\mathbb{P}_2 E \to L\mathbb{P}_2 A \to A$. By Corollary 1.4, $\pi_*(L\mathbb{P}_p(\Sigma^c E))$ is odd. □

Now we can explain the construction of $\mathbb{T}$.

Construction 1.6. We consider the following diagram.

$$\begin{array}{ccccc}
h\mathrm{Mod}^{\mathrm{ff}}_E & \xrightarrow{j} & h\mathrm{Mod}_E & \xrightarrow{\mathbb{T}_m} & \mathrm{Mod}_{E_*} \\
\pi_* \downarrow & & \pi_* \downarrow & \nearrow_{\mathbb{T}_m} & \\
\mathrm{Mod}^{\mathrm{ff}}_{E_*} & \xrightarrow{i} & \mathrm{Mod}_{E_*} & &
\end{array}$$

(with $\pi_*(L\mathbb{P}_m(-))$ the composite $h\mathrm{Mod}^{\mathrm{ff}}_E \to \mathrm{Mod}_{E_*}$)

where$\mathbb{T}_m$ is defined as the left Kan extension of $\pi_*(L\mathbb{P}_m(-))$ along $i\circ\pi_*(-)$, and $\tilde{\mathbb{T}}_m$ is defined as the composition. By universal property of the Kan extension, $\tilde{\mathbb{T}}_m$ is identified with the Left Kan extension of $\pi_*(L\mathbb{P}_m(-))$ along j. Therefore, we have a comparison transformation

$$\alpha_m\colon \mathbb{T}_m\pi_*(-) \to \pi_*(L\mathbb{P}_m(-))$$

which restricts to an equivalence on $h\mathrm{Mod}^{\mathrm{ff}}_{E_*}$.

We set $\mathbb{T} := \bigoplus \mathbb{T}_m$. Then α_m together with coproduct comparison give a transformation $\alpha\colon \mathbb{T}(-) \to \pi_*(L\mathbb{P}(-))$. Some properties of $\mathbb{T}$ and α are listed as follows.

Proposition 1.7. *Let $\mathbb{T}$ be as in Construction 1.6. Then we have:*

(i) $\mathbb{T}$ commutes with filtered colimits and reflective coequalizers.
(ii) $\mathbb{T}$ is strong symmetric monoidal, in the sense that we have $\mathbb{T}(M\oplus N)\cong\mathbb{T}(M)\otimes\mathbb{T}(N)$. As a result, for any $\mathbb{T}$-algebra M, M admits a commutative E_*-algebra structure given by

$$M\otimes M \to \mathbb{T}(M)\otimes\mathbb{T}(M)\cong\mathbb{T}(M\oplus M)\to\mathbb{T}(M)\to M$$

(iii) $\mathbb{T}$ is almost the best approximation. That means, for any flat E-module M, the comparison

$$\alpha\colon \mathbb{T}(M)\to\pi_*(L\mathbb{P}(-))$$

inherits $\pi_*(L\mathbb{P}(-))$ as the $\mathfrak{m}$-completion of $\mathbb{T}(M)$.

Therefore, there is a forgetful functor $U\colon \mathrm{Alg}_{\mathbb{T}}\to\mathrm{Alg}_{E_*}$. The forgetful functor satisfies a lot of good properties. Moreover, it is a standard example of a plethyistic functor.

Definition 1.8. Let $(\mathcal{C},\otimes,\mathbb{1})$ be a tensored Abelian category. A functor $U\colon\mathcal{D}\to\mathrm{CAlg}(\mathcal{C})$ is *plethyistic* if:

(1) U reflect isomorphisms.
(2) U admits both a left adjoint and a right adjoint.

A formal argument shows that a plethyistic functor U is both monadic and comonadic. Suppose we take F and W to be its left and right adjoint, then $U\circ F$ and $U\circ W$ becomes the associated monad and comonad, such that $\mathcal{D}$ is identified with the category of $U\circ F$-algebra and $U\circ W$-coalgebra.

Corollary 1.9. *The forgetful functor $U\colon\mathrm{Alg}_{\mathbb{T}}\to\mathrm{Alg}_{E_*}$ is plethyistic.*

Before we proceed on, we spend a little time to explain the concept of weight decomposition of a monad.

Definition 1.10. Let $\mathbb{T}$ be a monad on a symmetric monoidal category. A weight decomposition of $\mathbb{T}$ is a decomposition $\mathbb{T} \simeq \bigoplus \mathbb{T}_m$ together with following data

$$\begin{array}{ccc} \mathbb{1} & \longrightarrow & \mathbb{T}_0(M) \\ & \searrow^{\iota} & \downarrow \\ & & \mathbb{T}(M) \end{array} \qquad \begin{array}{ccc} \mathbb{T}_m(M) \otimes \mathbb{T}_n(M) & \longrightarrow & \mathbb{T}_{m+n}(M) \\ \downarrow & & \downarrow \\ \mathbb{T}(M) \otimes \mathbb{T}(M) & \longrightarrow & \mathbb{T}(M) \end{array}$$

$$\begin{array}{ccc} M & \longrightarrow & \mathbb{T}_1(M) \\ & \searrow^{\eta} & \downarrow \\ & & \mathbb{T}(M) \end{array} \qquad \begin{array}{ccc} \mathbb{T}_l\mathbb{T}_k(M) & \longrightarrow & \mathbb{T}_{lk}(M) \\ \downarrow & & \downarrow \\ \mathbb{T}\mathbb{T}(M) & \longrightarrow & \mathbb{T}(M) \end{array}$$

It is clear that our algebraic approximation monad admits a weight decomposition by construction.

1.2. **Additive power operations.** We now explain how to use the algebraic approximation monad $\mathbb{T}$ to understand power operations. Using this new notations, we can define power operations in a purely algebraic fashion. We begin with the degree 0 case.

Definition 1.11. We define the degree 0 power operations of E as

$$P = \mathrm{Hom}_{\mathrm{Alg}_{\mathbb{T}}}(\mathbb{T}(E_*), \mathbb{T}(E_*)) \simeq \mathrm{Hom}_{\mathrm{Mod}_{E_*}}(\mathbb{T}(E_*), E_*) \simeq \mathbb{T}(E_0)$$

It parametrizes endomorphisms of the forgetful functor $U\colon \mathrm{Alg}_{\mathbb{T}} \to \mathrm{Set}$.

There are two products on P. One is given by the algebra structure on $\mathbb{T}(E_0)$ and the other is given by the composition of the endomorphisms of the forgetful functor. We denote them by $\times$ and $\circ$ respectively. There are also two coproducts on $\mathbb{T}(E_0)$. One is given by $\Delta\colon \mathbb{T}(E) \to \mathbb{T}(E \oplus E)$ and the other is given by $E \simeq \mathbb{T}_1(E) \otimes \mathbb{T}_1(E) \to \mathbb{T}(E) \otimes \mathbb{T}(E)$. We denote them by Δ and ψ respectively.

Definition 1.12. We define the degree 0 additive power operations of E as the subset $\Gamma \subset P$ that corresponds to the endomorphisms of U that factor through Ab.

The coproduct Δ on P corresponds to the product on the source. As a result, $\Gamma \subset P$ consists of those primitive elements. That means, $x \in \Gamma$ if and only if $\Delta(x) = 1 \otimes x + x \otimes 1$. If we translate this into topological language, Δ corresponds to $\Delta\colon E\{t\} \to E\{x, y\}$ that sends $t \to x + y$. Therefore, for any $\alpha \in E_*^\wedge(B\Sigma_m)$, it is sent by Δ to $1 \otimes \alpha + \alpha \otimes 1$ plus the elements in the image of transfer maps. We have a short exact sequence:

$$0 \to \Gamma[k] \to E_0^\wedge(B\Sigma_{p^k}) \xrightarrow{\mathrm{Tr}} \bigoplus_{0 \le i \le p^k} E_0^\wedge(B\Sigma_i \times \Sigma_{p^k - i})$$

Note that for $m \neq p^k$, the transfer map above is an isomorphism. So we have a decomposition $\Gamma = \bigoplus_k \Gamma[k]$.

Proposition 1.13 ([Str98])**.** *We have $\Gamma[k] \subset E_0^\wedge(B\Sigma_{p^k})$ to be a dirrect summand. As a result, $\Gamma[k]$ is a free E_0-module.*

Note that the product $\circ$ restricts to Γ and we have $\Gamma[l] \circ \Gamma[k] \subset \Gamma[l + k]$. Moreover, we have the following:

Proposition 1.14. *The coproduct ψ and its unit restricts to Γ. As a result, Γ becomes a Hopf algebra under $\circ$ and ψ.*

Example 1.15. We consider the ht $= 1$ case. In this case, $\Gamma \cong \mathrm{W}(k)\langle\psi_p\rangle$, where ψ^p is the Adams operation and it satisfies:

$$\psi^p \cdot a = a^\sigma \cdot \psi^p$$

where $a \mapsto a^\sigma$ is the Frobenius lift. Note that unless $k = \mathbb{F}_p$, $\mathrm{W}(k)$ is not in the center of Γ.

Definition 1.16. Let Alg_Γ be the category of commutative Γ-algebras. Then we define U_1 to be the forgetful functor $U_1 \colon \mathrm{Alg}^*_{\mathbb{T}} \to \mathrm{Alg}_\Gamma$ that sends a $\mathbb{T}$-algebra B to its underlying algebra, with Γ-action given on the source of $B \cong \mathrm{Hom}_{\mathrm{Alg}_{\mathbb{T}}}(P, B)$.

We can generalize all the above constructions to arbitrary degrees. We denote $\omega^{q/2} = \mathbb{T}(\Sigma^q E)$. So we can define the degree i power operation as $P^i = \mathrm{Hom}_{\mathrm{Alg}_{\mathbb{T}}}(\omega^{q/2}, \mathbb{T}(E_*))$ and similarly define the degree i additive power operation Γ^i. In this case, a similar sequence as Proposition 1.13 shows up. We can identify $\Gamma^i[k] = \mathrm{Hom}_{\mathrm{Mod}_{E_*}}(E_0^\wedge(S^{-i}), K^i[k])$ and $K^i[k]$ fits into a short exact sequence

$$0 \to K^i[k] \to E_0^\wedge(B\Sigma_{p^k}^{-i\rho}) \xrightarrow{\mathrm{Tr}} \bigoplus_{0 \le i \le p^k} E_0^\wedge((B\Sigma_i \times \Sigma_{p^k - i})^{-i\rho})$$

where ρ is the permutation representation of Σ_{p^k}.

Using the monad structure of $\mathbb{T}$, $\{\Gamma^i, i \in \mathbb{Z}\}$ together forms a graded ring Γ^*. Putting together, we have a chain of forgetful functors

$$\mathrm{Alg}^*_{\mathbb{T}} \xrightarrow{U_1} \mathrm{Alg}^*_\Gamma \xrightarrow{U_2} \mathrm{Alg}_{E_*}$$

2. Congruence criteria

After we constructed the algebraic approximation functor $\mathbb{T}$ and the additive power operations Γ^*, a natural question to ask is: what is the distance between $\mathrm{Alg}^*_{\mathbb{T}}$ and the Alg^*_Γ, for the former contains the data of power operations and the latter contains those additive ones. We have some results. First, by some graded Hopf algebra theories, one can show the following:

Proposition 2.1. *When restricting to rational objects, the forgetful functor $U_1 \colon \mathrm{Alg}^*_{\mathbb{T}} \to \mathrm{Alg}^*_\Gamma$ induces an equivalence*

$$\mathrm{Alg}^*_{\mathbb{T},\mathbb{Q}} \simeq \mathrm{Alg}^*_{\Gamma,\mathbb{Q}}$$

In general, we have the *congruence criteria*, which is the main object we are going to discuss in this section.

We begin with the settings of the congruence criteria.

Notation 2.2. We use J to denote the augmented ideal of $E^0(B\Sigma_m) \to E_0$ and use I to denote the transfer ideal in $E^0(B\Sigma_m)$, that is, the image of $\bigoplus_i E^0(B\Sigma_i \times \Sigma_{m-i}) \xrightarrow{\mathrm{Tr}} E^0(B\Sigma_m)$.

We can consider the following fiber of the finite cover $B\Sigma_i \times \Sigma_{m-1} \to B\Sigma_m$:

$$\begin{array}{ccc} \Sigma_m/\Sigma_i \times \Sigma_{m-i} & \longrightarrow & B(\Sigma_i \times \Sigma_{m-i}) \\ \downarrow & & \downarrow \\ * & \longrightarrow & B\Sigma_m \end{array}$$

We can apply the double coset formula or the proper base change for $E^0(-)$. We have:

$$\begin{array}{ccc} E^0(\Sigma_m/\Sigma_i \times \Sigma_{m-i}) & \xleftarrow{j^*} & E^0(B\Sigma_i \times \Sigma_{m-i}) \\ \downarrow{\scriptstyle \mathrm{Tr}'} & & \downarrow{\scriptstyle \mathrm{Tr}} \\ E^0 & \xleftarrow{i^*} & E^0(B\Sigma_m) \end{array}$$

Therefore, $i^*(I) \subset E^0$ is exactly the ideal generated by the binomial coefficients $\binom{m}{i}$, which contains a unit unless $m = p^k$. In this case, $i^*(I) = (p)$. Especially, we have the following:

Proposition 2.3. *When $m = p$, the augmented map $E^0(B\Sigma_p) \to E^0$ restricts to an equivalence $I \cong pE^0$. Therefore, we have a pullback square:*

$$\begin{array}{ccc} E^0(B\Sigma_p) & \xrightarrow{\pi^*} & E^0(B\Sigma_p)/I \\ {\scriptstyle i^*}\downarrow & \lrcorner & \downarrow{\scriptstyle \sigma^*} \\ E^0 & \xrightarrow[\pi^*]{} & E^0/(p) \end{array}$$

Proof. In this case, I is a rank 1 free E_0-module, so the surjection $I \to pE_0$ is an isomorphism. □

Var the duality $\mathrm{Hom}(E^0(B\Sigma_p)/I, E^0/p) \cong \Gamma[1]/p$, we can choose an element $\sigma \in \Gamma[1]/p$ corresponding to $\sigma^* \colon E^0(B\Sigma_p)/I \to E^0/p$. We also use $\tilde{\sigma} \in \Gamma[1]$ to denote a lift of σ and call it a *Frobenius element*.

Let M be a E_* module. We define

$$u \colon \Gamma[1] \otimes M \to \mathbb{T}_p M \quad v^p \colon \mathrm{Sym}^p(M) \to \mathbb{T}_p M$$

corresponding to the Γ-action map and p-th power map respectively.

Proposition 2.4. *The Frobenius element fits into a pushout square:*

$$\begin{array}{ccc} E_0 & \xrightarrow{p} & E_0 \\ {\scriptstyle (\tilde{\sigma}, -\mathrm{id})}\downarrow & \ulcorner & \downarrow \\ \Gamma[1] \oplus E_0 & \xrightarrow{(u,v)} & \mathbb{T}_p(E_0) \end{array}$$

Proof. By Proposition 2.3, we have the following map between short exact sequences:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & E^0(B\Sigma_p) & \xrightarrow{(\pi^*, i^*)} & E^0(B\Sigma_p)/I \oplus E^0 & \xrightarrow{(\sigma^*, \pi^*)} & E^0/(p) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow{\scriptstyle (\sigma^*, -\mathrm{id})} & & \downarrow{\scriptstyle \cong} & & \\ 0 & \longrightarrow & E^0 & \xrightarrow{p} & E^0 & \longrightarrow & E^0/(p) & \longrightarrow & 0 \end{array}$$

Applying $\mathrm{Ext}^*_{E_0}(-, E_0)$ to the left square, we obtain the desired pushout square. □

We can now explain the congruence criteria.

Theorem 2.5 (Congruence criteria)**.** *Let $B \in \mathrm{Alg}^*_\Gamma$ be a torsion free object. Then B admits a $\mathbb{T}$-algebra structure if and only if it satisfies the congruence criteria. That is, for every $x \in B_0$, we have $\tilde{\sigma} x = x^p \bmod p$.*

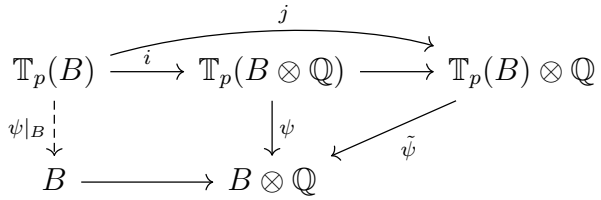
Proof. The only if part is obvious. Since suppose B itself is a $\mathbb{T}$-algebra, then $x \in B_0$ is represented by some $f\colon \mathbb{T}(E_0) \to B$. Therefore, $\tilde{\sigma}x - x^p$ is represented by a map of E_0 modules:

$$E_0 \xrightarrow{(\tilde{\sigma},-\mathrm{id})} \Gamma[1] \oplus E_0 \xrightarrow{(u,v)} \mathbb{T}_p(E_0) \xrightarrow{f} B$$

By Proposition 2.4, this map must factor through the multiplication by p. This proves the only if part.

Conversely, suppose that $B \in \mathrm{Alg}^*_\Gamma$ satisfies the congruence criteria, our strategy is to consider the embedding $B \hookrightarrow B \otimes \mathbb{Q}$. $B \otimes \mathbb{Q}$ always admits a $\mathbb{T}$-algebra structure $\psi\colon \mathbb{T}(B \otimes \mathbb{Q}) \to B \otimes Q$ restricts to $\mathbb{T}(B)$. We first consider the weight p case.

Consider the following diagram. ψ factors through $\mathbb{T}(B) \otimes \mathbb{Q}$ and by monadic structure, $\tilde{\psi}$ sends $B \hookrightarrow \mathbb{T}(B) \otimes \mathbb{Q}$ to B. The upshot is the image of $j\colon \mathbb{T}_p(B) \to \mathbb{T}_p(B) \otimes \mathbb{Q}$ is generated by $u(\Gamma[1] \otimes B)$, $v^p(\mathrm{Sym}^p(B))$ and $(\tilde{\sigma}x - x^p)/p$ for $x \in B^0$ (This can be shown after reducing to M is a shift of copy of E_0, which is essentially Proposition 2.4). The map $\tilde{\psi}$ takes the first and second types of generators to B since they are already in B, and the third is also in B by congruence criteria. So we have a restriction $\psi|_B$ that makes the diagram commute.



The following Lemma 2.7 and Lemma 2.8 shows that the argument in weight p case is already enough. □

In order to finish the above proof, that is, to generalize the argument to arbitrary weight, we need some new notations.

Definition 2.6. Let $\mathbb{T}$ be a monad on a symmetric monoidal category $\mathcal{C}$ with a weight decomposition, as defined in Definition 1.10. Then we say that m is a *regular weight* if for every $M \in \mathcal{C}$, the family

$$S = \{\delta_i\colon \mathbb{T}_i(M) \otimes \mathbb{T}_{m-i}(M) \to \mathbb{T}_m(M); \mu_d\colon \mathbb{T}_d\mathbb{T}_{m/d}(M) \to \mathbb{T}_m(M)\}$$

is an epimorphism family. That is, for every $f, g \in \mathcal{C}$, $f\delta_i = g\delta_i$ and $f\mu_d = g\mu_d$ implies $f = g$. We say that m is a *critical weight* if it is not a regular weight.

The behavior of a monad with a weight decomposition is somehow determined by its critical weight. More precisely, we have the following:

Lemma 2.7. *Let $\mathbb{T}$ be a monad with weight decomposition over a symmetric monoidal category $\mathcal{C}$. Let $\psi : \mathbb{T}(M) \to M$ be a $\mathbb{T}$-algebra and let $N \subset M$ be a subobject. Then ψ restricts to an algebra structure on N if and only if $\psi_m\colon \mathbb{T}_m(M) \to M$ restricts to N for every critical weight m.*

Proof. We prove this by induction on m. Suppose that $m \geq 2$ is not a critical weight, we need to show that ψ_m restricts to $\psi_m|_N\colon \mathbb{T}_m(N) \to N$. This is because we always have the

followings:

$$\begin{array}{ccccc} \mathbb{T}_i(N)\otimes\mathbb{T}_{m-i}(N) & \longrightarrow & \mathbb{T}_m(N) & \longrightarrow & \mathbb{T}_m(M) \\ \downarrow & & \vdots & & \downarrow{\scriptstyle \psi} \\ N\otimes N & \xrightarrow{m} & N & \longrightarrow & M \end{array}$$

$$\begin{array}{ccccc} \mathbb{T}_d\mathbb{T}_{m/d}(N) & \xrightarrow{\mu} & \mathbb{T}_m(N) & \longrightarrow & \mathbb{T}_m(M) \\ {\scriptstyle \mathbb{T}_d(\psi_{m/d})}\downarrow & & \vdots & & \downarrow{\scriptstyle \psi} \\ \mathbb{T}_d(N) & \xrightarrow{\psi} & N & \longrightarrow & M \end{array}$$

The dotted arrows exist because the larger squares commute, and it makes the right square commute because of the assumption that m is a regular weight. □

Now Theorem 2.5 follows from the following observation.

Lemma 2.8. *Let $\mathbb{T}$ be our algebraic approximation monad on Mod_{E_*}. Then the only critical weight for $\mathbb{T}$ is p.*

Proof. Since $\mathbb{T}$ commutes with reflective coequalizers and filtered colimits, it suffices to show that the family S in Definition 2.6 is an epimorphism family for M finite free. We will show that:

(1) In the case $p \nmid m$, choose k such that p^k is the largest p-power that is smaller than m, then
$$\delta\colon \mathbb{T}_{p^k}(M)\otimes\mathbb{T}_{m-p^k}(M)\to\mathbb{T}_m(M)$$
is surjection.

(2) In the case $p|m$, the map
$$\mu\colon \mathbb{T}_p(M)\mathbb{T}_{m/p}(M)\to\mathbb{T}_m(M)$$
is surjection.

Since we reduced to the case that M is finite free. Then we can choose $X\in\mathrm{Mod}_E$ such that $\pi_*(X)\cong M$. Then we need to show that $\pi_*L\mathbb{P}_{\Sigma_{p^r}\times\Sigma_{m-p^r}}(N)\to\pi_*L\mathbb{P}_m(N)$ and $\pi_*L\mathbb{P}_{\Sigma_p\wr\Sigma_{m/p}}(N)\to\pi_*L\mathbb{P}_m(N)$ is surjective. This follows from Lemma 1.3. □

3. Isogenies associated with power operations

In this section, we explain the relationship between isogenies and additive power operations. In this section, any vector bundle is complex vector bundle and we fix notations $\mathcal{O}_{\mathrm{univ}}=E_0$ and $\mathcal{O}_G=E_0[\![t]\!]=E^0(BS^1)$. We begin with some recollections of basic definitions.

Definition 3.1. Let G and G' be two formal groups. An isogeny is a formal group map $\phi\colon G\to G'$ which is finite and free.

Suppose the formal groups are over a complete local ring R, then any isogeny is represented by a map $f\colon R[\![t]\!]\to R[\![t]\!]$ such that $f(t)=ct^n$ after mod $(x^{n+1},\mathfrak{m})$. One can also think of isogenies as the maps that "behave very like Frobenius". In fact, we will see that in many cases the Frobenius are the only interesting isogenies.

Definition 3.2. Let $\hat{\mathcal{R}}$ denote the category of complete local rings and let Γ be a formal group over k.

(1) For $B \in \hat{\mathcal{R}}$, a *deformation of* Γ *to* B is a triple $(\mathbb{G}, i, \alpha)$, where $\mathbb{G}$ is a formal group over B, $i\colon B/\mathfrak{m} \to k$ such that after passing to special fiber we have an isomorphism $\alpha\colon \mathbb{G}_0 := \pi^*\mathbb{G} \cong i^*\Gamma$.
(2) Let $\phi\colon \Gamma \to \Gamma'$ be an isogeny of formal groups over k. Then a deformation of ϕ to B consists of a deformation $(\mathbb{G}, i, \alpha)$ of Γ to B and a deformation $(\mathbb{G}', i, \alpha')$ and an isogeny $q\colon \mathbb{G} \to \mathbb{G}'$, which fits into the following diagram:

$$\begin{array}{ccc} \mathbb{G}_0 & \xrightarrow{q_0} & \mathbb{G}_0' \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\alpha'} \\ i^*\Gamma & \xrightarrow{i^*\phi} & i^*\Gamma' \end{array}$$

We let $\mathrm{Def}_\Gamma(B)$ denote the groupoid of deformations of Γ to B. This defines a groupoid value presheaf on $\hat{\mathcal{R}}$, which is represented by the formal scheme $\mathrm{Spf}(E_0)$ by [LT66]. Let $\mathrm{Def}_q(-)$ denote the groupoid value presheaf defined by deformations of q. We will see that this is also represented by a formal scheme. Moreover, there is naturally two maps $s\colon \mathrm{Def}_\psi(-) \to \mathrm{Def}_\Gamma(-)$ and $t\colon \mathrm{Def}_\psi(-) \to \mathrm{Def}_{\Gamma'}(-)$, which corresponds to the source and target of an isogeny.

Our next aim is to study the deformation of isogenies more closely. We will need to recall more notations.

Definition 3.3. Let R be a complete local ring and G be a formal group over R. An *effective divisor* of degree d on G is an ideal $\mathcal{I}(D) \subset \mathcal{O}_G$ such that $\mathcal{O}_G/\mathcal{I}(D)$ is a free R-module of rank d. Alternatively, the ideal is generated by some monic $f(x)$ of degree d such that the non-leading coefficience are in $\mathfrak{m}$.

Definition 3.4. Let G be a formal group over a complete local ring R.

(1) A subgroup K of G is an effective divisor which is also a subgroup. That is, it respects the unit law and multiplications:

$$\begin{array}{ccccc} K \times K & \dashrightarrow & K & & \\ \downarrow & & \downarrow & \nwarrow & \\ G \times G & \longrightarrow & G & \longleftarrow & * \end{array}$$

Alternatively, an effective divisor K is a subgroup if and only if the defining ideal $\mathcal{I} = (f(x))$ is generated by $f(x)$ such that $f(x +_G y) \in (f(x), f(y)) \in R[\![x, y]\!]$ and $f(x) \in xR$.
(2) Let $G \to G'$ be an isogeny of formal groups, then the kernel K is defined by $\mathcal{O}_K := \mathcal{O}_G \otimes_{\mathcal{O}_{G'}} R$. It is indeed a subgroup.
(3) Let $K < G$ be a subgroup. Then the quotient $f\colon G \to G/K$ is defined as the equalizer

$$\mathcal{O}_{G/K} \xrightarrow{f^*} \mathcal{O}_G \underset{\pi_1^*}{\overset{\mu^*}{\rightrightarrows}} \mathcal{O}_G \otimes \mathcal{O}_K$$

It can be checked that this construction satisfies a lot of good properties. For example, G/K is a formal group, $G \to G/K$ is a categorical quotient and an isogeny.

The following proposition shows that, in many cases, the Frobenius are the only isogenies.

Proposition 3.5. *If G is a formal group over a field k of character p, then (up to isomorphism) there is exactly one subgroup of degree p^r, which is the kernel of the Frobenius* Frob^r.

Proof. In fact, in this case, it can be checked that there is only one divisor of degree p^r, which is generated by x^{p^r}. □

The relation between divisors and deformation of isogenies can be summarized as follows. It is not surprising and the proof can be checked directly.

Proposition 3.6. *Let* $\mathrm{Frob}^r\colon \Gamma \to \Gamma'$ *be the Frobenius isogeny of formal groups over k. Let B be a complete local ring and $X^r_\Gamma(B)$ denote the groupoid, whose objects consist of pairs (G, H), such that G is a deformation of Γ to B and $H < G$ is a subgroup of degree p^r. Then there is an isomorphism of presheaves* $\mathrm{Def}_{\mathrm{Frob}^r}(B) \cong X^r_\Gamma(B)$.

Proof. Suppose that q is a deformation of Frob^r, then we can associate it with its kernel, which is a subgroup of degree p^r. Conversely, suppose that we have a subgroup K of a deformation G which is of degree p^r, then after passing to its special fiber, everything is over a field, so by Proposition 3.5, the special fiber K_0 must be the kernel of the Frobenius. □

Now we are going to construct the isogeny associated with the total power operation. Before that, we still need some preparations. We consider a space X with even E-cohomology, the examples include $X = *$, $B\Sigma_m$ or BS_1.

We consider the finite cover $B(\Sigma_i \times \Sigma_{m-i}) \to B\Sigma_m$. If we have some vector bundle $W \to X \times B\Sigma_m$, we can pullback it to $B(\Sigma_i \times \Sigma_{m-i})$. Therefore we can apply the double coset formula or proper base change to the square

$$\begin{array}{ccc} W' & \longrightarrow & W \\ \downarrow & & \downarrow \\ X \times B(\Sigma_i \times \Sigma_{m-i}) & \longrightarrow & X \times B\Sigma_m \end{array}$$

We can do the same thing to the sphere bundle or Thom space. Therefore we have

$$\begin{array}{ccc} E^0(S(W')) & \xrightarrow{\mathrm{Tr}} & E^0(S(W)) \\ {\scriptstyle\pi^*}\uparrow & & \uparrow{\scriptstyle\pi^*} \\ E^0(X \times B(\Sigma_i \times \Sigma_{m-i})) & \xrightarrow{\mathrm{Tr}} & E^0(X \times B\Sigma_m) \end{array}$$

If we define $I \subset E^0(X \times B\Sigma_m)$ and $I' \subset E^0(S(W'))$ as the direct sum of images of transfers, the proper base change tells us that we have $\pi^*(I) \subset I'$. If the left vertical arrow is a surjection, then we have $\pi^*(I) = I'$. Similarly, we can define the transfer ideal $I'' \subset E^0((X \times B\Sigma_m)^W)$ and $z^*(I'') = I$, where $z\colon X \times B\Sigma_m \to (X \times B\Sigma_m)^W$ is the Gysin map.

Construction 3.7. We have the following diagram:

$$\begin{array}{ccccccc}
0 & & & & 0 & & 0 \\
\uparrow & & & & \uparrow & & \uparrow \\
E^0(S(L)) & \dashrightarrow & E^0(S(L^m)) & \longrightarrow & E^0(L\otimes V_m) & \longrightarrow & E^0(L\otimes V_m)/I' \\
{\scriptstyle \pi^*}\uparrow & & \uparrow & & {\scriptstyle \pi^*}\uparrow & & {\scriptstyle \pi^*}\uparrow \\
E^0(BS^1) & \xrightarrow{P_m} & E^0(BS^1 \wr \Sigma_m) & \longrightarrow & E^0(B(S^1\times\Sigma_m)) & \longrightarrow & E^0(B(S^1\times\Sigma_m))/I \\
{\scriptstyle z^*}\uparrow & & \uparrow & & {\scriptstyle z^*}\uparrow & & {\scriptstyle z^*}\uparrow \\
E^0(BS^{1^L}) & \xrightarrow{P_m} & E^0((BS^1\wr\Sigma_m)^{L_m}) & \longrightarrow & E^0(B(S^1\times\Sigma_m)^{L\otimes V_m}) & \longrightarrow & E^0(B(S^1\times\Sigma_m)^{L\otimes V_m})/I'' \\
\uparrow & & & & \uparrow & & \\
0 & & & & 0 & &
\end{array}$$

Here L is the universal line bundle over BS^1, V_m is the permutative representation of Σ_m and P_m is the total power operation. The red arrow is to emphasize that it is not a ring map since P_m is not additive in general. However, the composition of each row is indeed a ring map, since the only obstruction for total power operations to become additive is exactly the transfer ideal.

The first three columns are Gysin sequences. Since everything is even, the first and third columns are short exact. The last column is also exact because of the above discussions and the right upper square is a pushout square. Note that the image of z^* is generated by the image of the Thom class. Since the lower row sends any trivialization to trivialization, it sends the image of z^* to the image of z^*. Therefore the dotted arrow exists.

We define $\mathcal{O}_{\text{iso}} := E^0(B\Sigma_m)/I$. The image of z^* defines a divisor on $E^0(B(S^1\times\Sigma_m))/I$, we denote $\mathcal{O}_D := E^0(L\otimes V_m)/I'$. We use $s\colon \mathcal{O}_{\text{univ}} \to \mathcal{O}_{\text{iso}}$ to denote the algebra map. Therefore we have the following pushout square:

$$\begin{array}{ccc}
\mathcal{O}_G & \longrightarrow & \mathcal{O}_G \otimes^s_{\mathcal{O}_{\text{univ}}} \mathcal{O}_{\text{iso}} \\
\downarrow & \ulcorner & \downarrow \\
\mathcal{O}_{\text{univ}} & \longrightarrow & \mathcal{O}_D
\end{array}$$

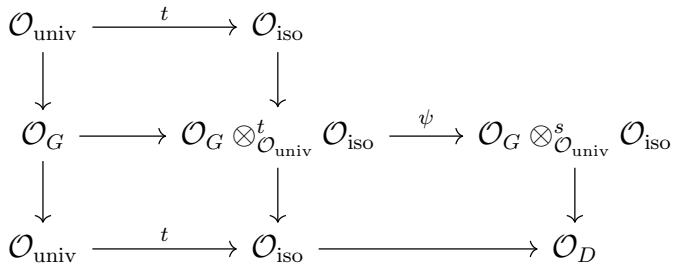
We consider the projection $BS^1 \to *$, which gives

$$\begin{array}{ccccc}
E^0(BS^1) & \xrightarrow{P_m} & E^0(B(S^1\times\Sigma_m)) & \longrightarrow & E^0(B(S^1\times\Sigma_m))/I \\
\uparrow & & \uparrow & & \uparrow \\
E^0 & \xrightarrow{P_m} & E^0(B\Sigma_m) & \longrightarrow & E^0(B\Sigma_m)/J
\end{array}$$

We let $t\colon \mathcal{O}_{\text{univ}} \to \mathcal{O}_{\text{iso}}$ denote the map given by total additive power operation. This gives another commutative square:

$$\begin{array}{ccc}
\mathcal{O}_{\text{univ}} & \xrightarrow{t} & \mathcal{O}_{\text{iso}} \\
\downarrow & & \downarrow \\
\mathcal{O}_G & \longrightarrow & \mathcal{O}_G \otimes^s_{\mathcal{O}_{\text{univ}}} \mathcal{O}_{\text{iso}}
\end{array}$$

Putting together, we obtain a diagram that contains the comparison map ψ:



where every square is a pushout square. We extract the map$\psi\colon \mathcal{O}_G \otimes^t_{\mathcal{O}_{\text{univ}}} \mathcal{O}_{\text{iso}} \to \mathcal{O}_G \otimes^s_{\mathcal{O}_{\text{univ}}} \mathcal{O}_{\text{iso}}$. If we translate this into the language of formal groups, we obtain a formal group map

$$\Psi\colon s^*G \to t^*G$$

where G is the universal formal group over $\mathcal{O}_{\text{univ}}$.

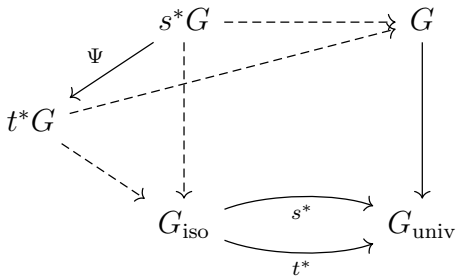
Proposition 3.8. *Let Ψ be constructed as above and let $m = p^r$, then:*

(1) Ψ is a formal group map.
(2) Ψ is an isogeny of degree m.
(3) Ψ is a deformation of $\mathrm{Frob}^r\colon G \to G^{(r)}$.

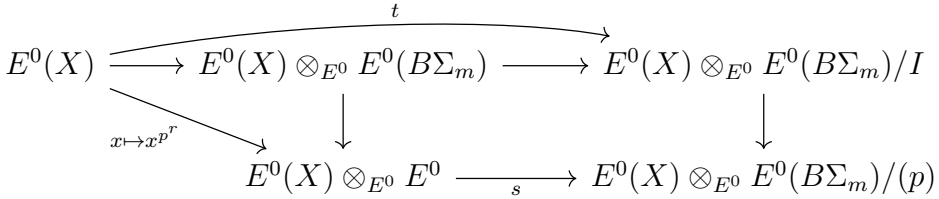
Proof. Note that the whole diagram in Construction 3.7 is functorial with respect to the term BS^1. Therefore, after replacing BS^1 by $BS^1 \times BS^1$ and consider the (co)multiplication, we obtain the first claim.

By construction, $\mathcal{O}_D$ is a finite free $\mathcal{O}_{\text{univ}}$-module of rank m. By the pushout diagram above, this implies that $\mathcal{O}_G \otimes^s_{\mathcal{O}_{\text{univ}}} \mathcal{O}_{\text{iso}}$ is a finite free $\mathcal{O}_G \otimes^t_{\mathcal{O}_{\text{univ}}} \mathcal{O}_{\text{iso}}$-module of the same rank, proving the second claim.

Now we have the following picture:



The claim that Ψ is a deformation of Frob^r reduces to the claim that $t^* = \mathrm{Frob}^r \circ s^*$ after passing to special fibers. It suffices to show that on the structure ring level, t coincides with $s \circ \mathrm{Frob}^r$ after mod p. Consider the map σ^* in Proposition 2.3 and the following commutative square finishes the proof.



□

We finish this section with some further remarks.

Remark 3.9. The isogeny in our construction is universal. In fact, it is shown in [AHS04, Proposition 12.12] and [Str98, Theorem 9.2] that we have $\mathrm{Def}_{\mathrm{Frob}^r}(-) \simeq \mathrm{Spf}(E^0(B\Sigma_m))$. That means, the constructed deformation Ψ is the universal one and this explains the notation $\mathcal{O}_{\mathrm{iso}}$. Moreover, the source and target map corresponds to $s, t\colon \mathcal{O}_{\mathrm{univ}} \to \mathcal{O}_{\mathrm{iso}}$, which represents the algebra map and total additive power operations respectively. The proof idea in [Str98] is by translating the language of deformation into the language of divisors. There is a universal divisor $\mathcal{O}_{\mathrm{Div}^r}$ that classifies all the divisors of degree p^r var pullback. The pullback of $\mathcal{O}_D$ on $\mathcal{O}_{\mathrm{iso}}$ defines a divisor over G_{iso}, hence a map $\mathcal{O}_{\mathrm{div}^r} \to \mathcal{O}_{\mathrm{iso}}$. One can show that this is an isomorphism by counting dimensions.

Remark 3.10. In [AHS04], a more general picture is shown. We know that every even periodic cohomology theory gives a formal group, and those even periodic cohomology theories with $\mathbb{H}_\infty$ structures corresponds those formal groups with descent data. Roughly, a descent data functorially assigns each finite abelian group with an isogeny, while in our talk we only cares the cyclic groups of order p^r.

4. Even freeness of $E^*(B\Sigma_m)$

We now briefly explain the even freeness of $E^*(B\Sigma_m)$. It is shown in [GS99] that this cohomology is finitely generated. In this section we will show the following:

Theorem 4.1. *The cohomology $E^*(B\Sigma_m)$ is even and free.*

Our tool to prove Theorem 4.1 is to break the Morava E-theory down to pieces and reduce to the Morava K-theory. More precisely, we have:

Lemma 4.2. *Let X be a spectrum such that $E^*(X)$ is finitely generated and $K^*(X)$ is even. Then $E^*(X)$ is even and free.*

Proof. We let $\mathfrak{m}_k$ denote the ideal $(v_0, \cdots, v_{k-1}) \subset E_0$ and consider the spectrum $E/\mathfrak{m}_k$. Especially $E/\mathfrak{m}_n \simeq K$ and $E/\mathfrak{m}_0 \simeq E$. We have a cofiber sequence

$$E/\mathfrak{m}_k \xrightarrow{v_k} E/\mathfrak{m}_k \to E/\mathfrak{m}_{k+1}$$

and after apply X, we obtain a short exact sequence

$$0 \to (E/\mathfrak{m}_k)^*(X)/v_k \to (E/\mathfrak{m}_{k+1})^*(X) \to \mathrm{ann}(v_k, (E/\mathfrak{m}_k)^*(\Sigma^{-1}X)) \to 0$$

By induction, each $(E/\mathfrak{m}_k)^*(X)$ is finitely generated. $(E/\mathfrak{m}_n)^*(X)$ is even and free. Suppose that $(E/\mathfrak{m}_{k+1})^*(X)$ is even and free, then from the short exact sequence we see that the odd part of $(E/\mathfrak{m}_k)^*(X)/v_k$ vanishes, so the odd part of $(E/\mathfrak{m}_k)^*(X)$ vanishes by Nakayama. The short exact sequence also shows that $\mathrm{ann}(v_k, (E/\mathfrak{m}_k)^{\mathrm{even}}(X))$ vanishes, therefore v_k acts regularly on $(E/\mathfrak{m}_k)^*(X)$. We have seen that $(E/\mathfrak{m}_k)^*(X)/v_k \subset (E/\mathfrak{m}_{k+1})^*(X)$ is free, so by regularity $(E/\mathfrak{m}_k)^*(X)$ is also free. □

For K-cohomology, we can compute everything through the Atiyah- Hurzebruch spectral sequence and its variance.

Lemma 4.3. *Let $W_m := C_p \wr \cdots \wr C_p$ denote the m times wreath of C_p over itself. Then $K^*(BW_m)$ is even.*

Proof. We argue this by induction. Suppose that this is true for BW_m, we can choose a basis $\{e_1, \cdots, e_d\}$ for the vector space $K^0(BW_m)$. Consider the following spectral sequence,

which is essentially the Atiyah-Hurzebruch spectral sequence by replacing sphere spectrum by BW_m:

$$E_2^{p,q} = \mathrm{H}^p(C_p; K^q(BW_n)^{\otimes p}) \Rightarrow K^{p+q}(BW_{n+1})$$

The generators of the second page are $z \in \mathrm{H}^1(C_p; \mathbb{F}_p)$, its Bockstein $y = \beta z$ and $e_I = e_{i_1} \otimes \cdots \otimes e_{i_p}$ for any combination I of the basis. We must figure out the differentials. First, e_I must be permanent cycles. This is because we can always find elements in $K^*(BW_{n+1})$ hitting these elements. For example, we can choose $\mathrm{Tr}_{W_m^p}^{W_{m+1}} e_I$ that hits e_I.

For the other generators, note that our spectral sequence is a module over the Atiyah-Hirzebruch spectral sequence

$$\mathrm{H}^p(C_p; K^q) \Rightarrow K^{p+q}(BC_p) \cong P[x]/x^{p^n}$$

while here the only none-trivial differential for z and y is $d_{p^n-1}(z) = y^{p^n}$. Combining together, we can eventually compute the spectral sequence and obtain the even freeness of $K^*(BW_{m+1})$, as shown in [Str98, Proposition 3.4]. □

Proof of Theorem 4.1. The transfer arguments (or more precisely, Lemma 1.3) allow us to reduce to show for p-Sylow subgroup of Σ_m, which reduces to show that $E^*(BW_m)$ is even free. By Lemma 4.2, it suffices to show that $K^*(BW_m)$ is even, which follows from Lemma 4.3. □

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